

Different Uses of Aggregation (Research & Reflect)

1. What is the difference between GROUP BY and ORDER BY?

- **GROUP BY** is used to **group rows that have the same values** in specified columns to perform **aggregate functions** (like **SUM()**, **AVG()**, **COUNT()**).
- **ORDER BY** is used to **sort the final output** of a query by one or more columns (ascending or descending).
- **GROUP BY** is about **combining data**, while **ORDER BY** is about **organizing data for display**.

2. Why do we use HAVING instead of WHERE when filtering aggregate results?

- **WHERE** filters **individual rows before aggregation**.
- **HAVING** filters **groups after aggregation** (i.e., it applies to the results of **GROUP BY**).
- We **cannot use aggregate functions** (like **SUM()**, **AVG()**) in **WHERE**, but We **can** in **HAVING**.

3. What are common beginner mistakes when writing aggregation queries?

- Using **WHERE** instead of **HAVING** for aggregated conditions.
- Selecting columns in **SELECT** that are **not included in GROUP BY** and not wrapped in an aggregate function.
- Forgetting to **alias aggregate results**, making output hard to read.
- Not understanding that **GROUP BY** reduces rows — you can't get row-level detail afterward.

4. When would you use COUNT(DISTINCT ...), AVG(...), and SUM(...) together?

In **reports or dashboards** that summarize key metrics, like:

- Total revenue (**SUM(Sales)**),
- Average revenue per order (**AVG(Sales)**),
- Number of unique customers (**COUNT(DISTINCT CustomerID)**).

Useful for **summarizing grouped data** by categories like department, region, or month.

5. How does GROUP BY affect query performance, and how can indexes help?

GROUP BY can be **computationally expensive** on large datasets because it requires sorting or hashing.

Performance improves when:

- **Indexes exist** on the columns being grouped.
- The database engine can **use indexed columns** to quickly locate and group rows.

Composite indexes or covering indexes are especially useful for improving **GROUP BY** efficiency.